

Molecular Docking Programs

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1 Introduction

As structure-based drug design developed into a mature science, molecular docking got better with every step. The method depends on how well the conformational search algorithm can find the bioactive conformation(s). However, the search algorithm by itself does not know what a bioactive conformation(s) looks like. By definition, bioactive conformation is the atomic arrangement of a molecule that is suitable for binding to its respective target. It either inhibits/antagonises or activates it. There is always a possibility that one ligand may modulate more than one target, in this case, the bioactive conformation may be different. However, the bioactive conformation is always one of the plausible low-energy states, not necessarily the global minimum.

Now that we have established that there are several possible bioactive conformations for a ligand depending upon its binding target, the conformational search algorithm alone is unable to identify it, but it can find it. The scoring function is responsible for scoring and ranking the conformations from the search algorithm. The chapter on scoring functions will treat this part in depth. Remember that the conformational search tool searches for the bioactive conformation within the receptor binding site; therefore, many low-energy conformations that are observed in the gas phase or solvent may not be sampled due to clashes with the receptor atoms. This is one of the ways by which docking programs implement a preliminary "bump-check" that can filter out conformations that show clashes with the receptor atoms. This way, more intense calculations are performed on fewer conformations. This improves the speed of docking.

With significant improvements in computing power, and easy of computing led to development of several molecular docking programs. Understandably, it is impossible to track and understand every bit of the development. Therefore, to assist, particularly, the beginners, this chapter will describe few

popular molecular docking programs. Care has been taken to include both commercially licensed programs and open-source programs.

There are a plethora of docking programs available (**Table 1**), some with commercial licenses and others with open-source licenses. Knowing every program to make a choice that best suits you is almost always impossible. However, this is also not necessary as there exists ample literature reporting the use of many of the tools. Irrespective of the choice of the docking program, the general steps will be same; for example, preparation of the receptor, defining the binding site as search space, preparation of ligands, specifying the docking parameters, validating the docking procedure, performing the docking calculations, and analysis and dissemination of the results. In a nutshell, the general principles remain unchanged; however, there are differences in ligand format, docking methods, and scoring algorithms that are well-documented in the user manual. We strongly recommend that, before using any docking algorithm, the user manual must be consulted to understand each parameter’s importance. The developers present their report on several validation studies as part of the software development, which is a great resource to understand the effect of each parameter on the outcome. A literature survey of the program’s usage will also be covered in the user manual.

Table 1: Non-exhaustive list of common molecular docking software

Software	License type	Scoring Function
AutoDock	Open Source	Force field
AutoDock-Vina	Open Source	Empirical
DOCK	Open Source (Academic)	Force field
GOLD	Commercial	Multiple options
FlexX	Commercial	Empirical
Glide	Commercial	Empirical
FLEXX	Commercial	Multiple options
SURFLEX	Commercial	Multiple options

2 Molecular Docking Programs:

The following sections will describe the working of the commonly available docking program (See Table 1)

2.1 DOCK

The DOCK program is considered the first tool developed for molecular docking in 1982. It was developed by the University of California, San Francisco (UCSF) in the group of Irwin Kuntz. It aims to predict the most favourable binding configurations of small molecules (ligands) within the active sites of macromolecular targets, such as proteins. This computational resource simulates molecular interactions to identify low-energy binding poses, making it an essential asset for drug discovery and structure-based design initiatives. Created to tackle rigid-body docking, DOCK utilises a geometric matching algorithm that aligns the ligand with a negative representation of the receptor’s binding pocket. Over the years, its functionality has evolved to incorporate features such as force-field-based scoring, real-time optimisation, and algorithms for flexible ligand docking. The program employs scoring functions to assess and rank potential binding poses according to their predicted binding affinities. It accommodates various input file formats, including PDB, MOL2, and SDF, and is extensively utilised for virtual screening and lead optimisation in drug development. DOCK continues to be a fundamental tool in computational chemistry, renowned for its efficiency in exploring molecular interactions and its ability to adapt to new advancements in docking techniques.

2.1.1 Methodology

The docking method described in the sources involves a geometric approach to explore feasible alignments of ligands and receptors of known structure. We describe the process as shown in the original publication as follows:

1. *Representation*: The receptor and ligand structures are represented using spheres to fill pockets and grooves on the receptor’s surface and the space occupied by the ligand. These spheres are generated based on the molecular surface, which is defined using a probe sphere to determine contact and re-entrant points. The spheres are constructed such that each sphere touches the molecular surface at two points and has its center on the surface normal. The number of spheres is reduced by keeping only the smallest radius spheres, filtering based on the angle formed by the two surface points and the sphere centre, and by only retaining one sphere per atom. Receptor spheres are grouped into potential binding sites based on whether they overlap.
2. *Matching*: The ligand spheres are paired with the receptor spheres by comparing internal distances. A ligand sphere can be paired with a

receptor sphere if the internal distances of all the spheres in the ligand set match all the internal distances within the receptor set, within an error limit. The pairing process starts by systematically pairing each ligand sphere with each receptor sphere and then finding subsequent pairs based on whether their internal distances match. If the number of assignable pairs is less than four, the match is rejected. To reduce computation time, only ligand and receptor spheres whose centres are approximately the same distance from the centroid of each respective representation are accepted.

3. *Optimisation:* The suggested pairings are explored by rotating the ligand spheres onto the corresponding receptor spheres using a least-squares algorithm. The rotation/translation matrix generated from the spheres is applied to all ligand atoms. The optimisation procedure manipulates the ligand atom coordinates to reduce atom overlaps, ensure hydrogen-bonding partners, and place ligand atoms within receptor spheres. An overlap error is calculated based on the van der Waals' radii of the ligand and receptor atoms. The ligand is moved further into the receptor site by using the receptor sphere centre as a target for each ligand atom. Displacements are calculated for any ligand atom that violates the overlap constraint or is too far from a receptor polar atom, and the ligand is then moved to reduce these errors.

2.1.2 scoring:

The output is a set of ligand conformations/poses with scores based on the degree of overlap. Geometrically similar structures are grouped. The method employs a simplified interaction function that accounts for hard-sphere repulsions and hydrogen bonding. It is important to note that this method assumes rigid molecules and requires prior knowledge of the two structures to atomic detail. The calculated structures are intended as starting points for further refinement using molecular mechanics or interactive computer graphics. The approach also has independent applications, such as the location of binding sites and the design of new ligands.

2.2 GLIDE

One of the most popular methods for docking is Glide, which stands for **Grid-based Ligand Docking with Energetics**. This tool was developed by Schrödinger Inc. led by Prof. Richard Frienser and David Shaw from Columbia University and D. E. Shaw Research and Development, respectively[2].

Glide employs a systematic, hierarchical approach to ligand conformational filtering and pose selection, designed to efficiently explore the ligand's conformational space and its potential interactions within the receptor's active site while maintaining computational speed. This methodology involves a series of filters that progressively narrow the search space.

2.2.1 How does GLIDE search for ligand conformations?

The detailed description of each step in the GLIDE docking method is given below:

1. **Receptor Pre-computation and Grid Generation:** The protein receptor's shape and properties are precomputed and represented on grids, using different sets of fields for progressively more accurate scoring. A multigrid strategy is used, with finer resolution (down to 0.4 Å) closer to the protein's van der Waals surface, to ensure accuracy where ligand and protein contact. This pre-processing step creates an efficient and accurate representation of the rigid receptor environment, allowing rapid energy and gradient evaluations during subsequent ligand interaction calculations. The varying grid resolution optimises computational cost while maintaining precision in critical binding regions.
2. **Ligand Conformation Generation:** Glide first performs an exhaustive conformational search in the ligand's torsion-angle space to generate initial conformations. A heuristic screen rapidly eliminates high-energy conformers by evaluating torsional energy using a truncated OPLS-AA potential and applying an energy cut-off. Ligands are conceptually divided into a rigid "core" region and flexible "rotamer groups" attached by rotatable bonds. Each core conformation is enumerated, and then combined with all possible rotamer states, allowing thousands of effective conformations to be considered. This step aims for a comprehensive sampling of ligand flexibility, ensuring that relevant bioactive conformations are likely to be generated. The core/rotamer model and heuristic screening make the exhaustive search computationally feasible for even large and flexible ligands.
3. **Initial Screening of Ligand Poses:** For each generated ligand conformation, an exhaustive search of possible positions and orientations is performed within the active site. This multi-stage process includes:
 - (a) **Site Point Selection:** Placing the ligand centre at "site points" on a 2 Å grid if there's a good match between ligand and receptor surface distances.

- (b) Orientation Screening: Testing pre-specified orientations of the "ligand diameter" and rejecting those with excessive steric clashes.
- (c) Subset Scoring: Scoring only atoms capable of forming hydrogen bonds or metal interactions (a "subset test").
- (d) Full Interaction Scoring: Scoring all ligand-receptor interactions for promising orientations.
- (e) These tests use a discretised ChemScore version, rapidly identifying favourable hydrophobic, hydrogen-bonding, and metal-ligation interactions while penalising clashes. Approximately 5000 top greedy-scoring poses are then subjected to a rigid refinement allowing 1 Å Cartesian movement.

This is a rapid pre-screening stage that drastically reduces the conformational and positional search space. By using a fast, approximate scoring function ("greedy scoring"), it efficiently filters out unlikely poses, making more computationally intensive subsequent steps feasible. The exhaustive nature aims to avoid missing key regions of phase space.

4. Energy Minimization (SP Minimization): A select number of the best poses (typically 400) from the initial screening are subjected to energy minimization using precomputed OPLS-AA van der Waals and electrostatic grids. The minimization often begins on "smoothed" grids to avoid deep local minima, gradually transitioning to the full OPLS-AA non-bonded energy surface ("annealing"). This step accounts for rigid-body translations and rotations, and for flexible docking, also includes torsional motion around rotatable bonds. This step refines the rough poses into more energetically stable configurations within the receptor's force field. The annealing approach helps to overcome local energy barriers, leading to more accurate pose prediction.
5. Post-Docking Minimisation and Monte Carlo Refinement: The very best poses (typically 3 to 6) from the energy minimisation are subjected to an additional Monte Carlo procedure. This procedure explores nearby torsional minima, refining the orientation of peripheral groups and occasionally adjusting internal torsion angles. This final refinement step ensures that the most promising poses are thoroughly optimized, capturing subtle conformational adjustments that may be crucial for accurate binding mode prediction.
6. Pose Selection (Emodel): The ultimate selection of the best-docked

pose is made using Emodel, a composite energy score. Emodel combines the GlideScore (an expanded ChemScore-like empirical function), the ligand-receptor molecular mechanics interaction energy, and the ligand’s internal strain energy (for flexible docking). Emodel provides a robust and accurate criterion for selecting the most representative binding pose. By integrating multiple energy components, it offers a more balanced assessment of pose quality compared to using individual scoring functions alone.

2.2.2 How does GLIDE Score the docked conformations?

Glide employs several scoring functions, primarily GlideScore (with Standard Precision (SP) and Extra Precision (XP) versions) and Emodel, a composite score used for final pose selection. These functions integrate various physical and empirical terms to evaluate ligand binding affinity and rank poses. The general form of the GlideScore 2.5 SP function is given as:

$$\begin{aligned} \Delta G_{\text{bind}} = & C_{\text{lipo-lipo}} \sum f(r_{lr}) + \\ & C_{\text{hbond-neut-neut}} \sum g(\Delta r) h(\Delta R) + \\ & C_{\text{hbond-neut-charged}} \sum g(\Delta r) h(\Delta R) + \\ & C_{\text{hbond-charged-charged}} \sum g(\Delta r) h(\Delta R) + \\ & C_{\text{max-metal-ion}} \sum f(r_{lm}) + \\ & C_{\text{roth}} H_{\text{roth}} + C_{\text{polar-phob}} V_{\text{polar-phob}} + \\ & C_{\text{coul}} E_{\text{coul}} + C_{\text{vdW}} E_{\text{vdW}} + \text{solvation terms} \end{aligned} \tag{1}$$

Here are the key components of Glide’s scoring functions:

1. GlideScore (General Components - Shared between SP and XP)
 - (a) Lipophilic-Lipophilic Term: This term, similar to that in ChemScore, accounts for interactions between lipophilic atoms of the ligand and the receptor. It heuristically represents the displacement of water molecules from hydrophobic regions by lipophilic ligand atoms, which lowers the free energy via the hydrophobic effect.
 - (b) Hydrogen-Bonding Term: Based on ChemScore, this term is separated into components with different weights depending on the charge state of the donor and acceptor atoms: neutral-neutral,

neutral-charged, and charged-charged. In the optimized scoring function, neutral-neutral contributions are found to be the most stabilizing, while charged-charged are the least important. The quality of hydrogen bonds is evaluated based on geometric criteria (angles and distances).

- (c) Metal-Ligand Interaction Term: This term considers interactions with anionic acceptor atoms, recognising a strong preference for coordination of anionic ligand functionality to metal centres in metalloproteases. Glide 2.5 counts only the single best interaction if multiple metal ligations are found. The preference for anionic ligands is suppressed if the metal ion in the apo-protein is neutral.
 - (d) Coulomb and van der Waals (vdW) Interaction Energies: These molecular mechanics components contribute to GlideScore 2.5. To make the gas-phase Coulomb interaction energy a better predictor, Glide reduces the net ionic charge on formally charged groups (e.g., carboxylates, guanidiniums) by approximately 50%. And reduces the vdW interaction energies for the directly involved atoms. The full interaction energy is used only for anionic ligand-metal interactions.
 - (e) Solvation Terms / Desolvation Penalties: Glide 2.5 incorporates a solvation model to address the requirement that charged and polar groups of both the ligand and protein must be adequately solvated. Explicit water molecules are docked into high-scoring poses, and empirical scoring terms measure the exposure of various groups to these waters. Penalties are assessed if a polar or charged ligand or protein group is inadequately solvated or if water molecules are trapped in hydrophobic pockets. This "water-scoring" technology is made efficient by grid-based algorithms.
2. Extra Precision (XP) GlideScore Specific Novel Terms: The XP scoring function and docking protocol were developed to reproduce experimental binding affinities and yield quality enrichments, highlighting the importance of novel molecular recognition and water scoring in separating active and inactive ligands and avoiding false positives. These terms aim to identify specific structural motifs that provide large contributions to enhanced binding affinity.
- (a) Hydrophobic Enclosure (E_{hyd•enclosure}): This term rewards situations where groups of lipophilic ligand atoms are enclosed on opposite faces by lipophilic protein atoms. It goes beyond a sim-

ple atom-atom pair term, recognizing that enclosed hydrophobic micro-environments are particularly unfavourable for water molecules, leading to significant binding free energy gains upon water displacement. A minimum group size of three connected lipophilic ligand atoms is considered for this effect. The score is capped at 4.5 kcal/mol.

- (b) Special Neutral-Neutral Hydrogen-Bond Motifs (Ehb●nn●motif): These motifs receive additional binding affinity increments. They are identified in situations where a water molecule forming a hydrogen bond to the protein would have particular difficulty in making its complement of additional hydrogen bonds, especially when there are hydrophobic protein atoms on two faces. This is often crucial for high potency in pharmaceutical targets like kinases. The reward is 1.5 kcal/mol for a single bond to a ring in hydrophobic environments and 3.0 kcal/mol for a neutral pair in hydrophobic environments. The protein’s hydrogen-bond partner must have a limited number of waters in its first or second shell. Correlated hydrogen bonds between protein and ligand can also receive rewards if they are in a challenging hydrophobic environment.
- (c) Special Charged-Charged Hydrogen-Bond Motifs (Ehb●cc●motif): These terms signal enhanced binding affinity from charged-charged hydrogen bonds based on:
 - i. Number of waters surrounding the protein component of the salt bridge, as fully solvent-exposed charged groups are unlikely to participate in enhanced interactions.
 - ii. Number of charged-charged hydrogen bonds (monodentate, bidentate, or one ligand group to two different protein groups).
 - iii. Zwitterion ligands: For zwitterions, the long-range effects of solvation free energy are significantly reduced, making the formation of two salt bridges more favourable.
 - iv. Distinguishing between positive ligand/negative protein and vice versa.
 - v. Strength of the electrostatic field at the ligand.
 - vi. Rewards for various motifs range from 0.5 to 4.7 kcal/mol.
- (d) Pi-Stacking and Pi-Cation Interactions (EPI): These specialised terms, still under development, reward such interactions.
- (e) Empirical Correction for Smaller Ligands: An empirical correction

enhances the binding affinity of smaller ligands relative to larger ones.

3. Penalty Terms in XP GlideScore:

- (a) Desolvation Penalties (E_{desolv}): As described under solvation terms above, these specifically penalize inadequately solvated polar or charged groups and waters in hydrophobic pockets.
- (b) Ligand Strain Energy (Eligand•strain): This penalises poses with close internal contacts within the ligand. It is a challenging aspect due to the rigid-receptor approximation. Still, Glide includes functions to count intramolecular heavy atom contacts and evaluate penalties based on the size and location (e.g., peripheral vs. central) of contacting groups.

4. Emodel

- (a) Emodel is a composite energy score used for the ultimate selection of the best-docked pose.
- (b) It combines the GlideScore, the ligand-receptor molecular mechanics interaction energy, and the ligand’s internal strain energy (for flexible docking). This composite function is found to be superior in selecting the correct pose compared to using molecular mechanics energy or GlideScore alone.

2.3 AUTODOCK

Autodock or commonly known as AD4, offers an easy-to-set-up and run kind of environment for molecular docking experiments. Therefore, this becomes an easy tool for beginners wanting to learn molecular docking. The developers also separately provide all the tools/scripts, known as "MGLTools", necessary to set up chemical systems for molecular docking. It is, therefore, an automated computational procedure primarily designed to predict the bound conformations of small, flexible ligands to macromolecular targets of known structure. Being an open-source tool with source code available, various groups have expanded the functionality of AD4 to improve docking accuracy for specific and challenging problems. The method balances the need for a robust and physically relevant procedure with the requirement to keep computational demands at a feasible level. Its core operation combines efficient conformation searching with rapid energy evaluation. Here’s a detailed explanation of how AutoDock searches for ligand conformations within the receptor pocket:

1. Conformational Searching Methods AutoDock provides the user with the choice of many well-tested conformational search algorithms. Based on the problem at hand, and with little exploratory work, the user will be able to find out for themselves which of the search algorithms is more appropriate for them. This, however, could change from problem to problem because a different biomolecular system could be best suited for another search algorithm.

Here we explain each of the search algorithms available with AD4.

(a) Simulated Annealing (SA):

Simulated annealing is a concept more prevalent in engineering science, where malleability in metals/alloys is employed to manufacture cables for a vast variety of applications. For example, a long steel cable can be manufactured from a block of steel by heating then beating and then cooling. This cycle is repeated until the desired shape, size and length is achieved, which dictates the tensile strength of the steel cable. This heating-beating-cooling-heating cycle is simulated annealing. The heating increases the softness of the steel by making the atoms move apart by gaining kinetic energy at high temperatures. Thus, making the beating process easier and preventing any fracture or permanent deformation. The cooling stage traps the atoms in the beaten state. At the end of each cycle, the shape of the steel block progressively approaches the desired shape and size without losing the inherent properties of steel.

Similarly, in sampling the conformations of chemical/biological systems, the heating stage allows the sampling of high-energy, low-probability regions. While the cooling stage drops the system to the nearest low-energy stable conformation. This way, the search algorithm can explore the accessible potential energy space of the system.

- i. Initial Approach: Earlier versions of AutoDock (such as version 1.0) primarily used the Metropolis technique, also known as simulated annealing, for conformational and positional searching.
- ii. Mechanism: Here the conformational search is performed by making the molecule take a random walk in the space around the static target protein. At each step, small random displacements are applied to the ligand's rigid body translation, rotation, and each torsion angle. The interaction energy for

this new state is evaluated, and the new state is accepted if the energy is lower than the previous state. If the new energy is higher, the step is accepted probabilistically, depending on a user-defined "temperature". High temperatures allow sampling of states located at high energy in the potential energy surface. Therefore, if the temperature selected by the user is higher, then the probability of accepting high energy states increases. In a nutshell, such a criteria for accepting or rejecting conformations based on random sampling is based on the *Metropolis-Hashing algorithm*. The Probability distribution based on the sampling temperature must follow *Boltzmann distribution*.

- iii. Refinement: Once the sampling algorithm has identified conformation that show promising interaction energy. The next logical step should to refine these conformations in the presence of the receptor. Note, that the previous step, *i.e.* it is performed in the grid box whose vertices carry information about the likely interaction type with the receptor atoms-affinity maps. The binding conformation is refined as the simulation proceeds by gradually reducing the temperature. The simulation is broken into cycles at a constant temperature, and the temperature is reduced at the beginning of each new cycle. Each cycle typically begins with the trial molecule in the lowest energy conformation found in the previous cycle.
 - iv. Limitations: While SA performed well for ligands with roughly eight or fewer rotatable bonds, problems with more degrees of freedom rapidly became intractable. It could also sometimes fail to reproduce crystal structures in challenging cases.
- (b) Genetic Algorithms (GA) and Lamarckian Genetic Algorithm (LGA)
- GAs use principles from natural genetics and biological evolution. A ligand's arrangement (translation, orientation, conformation) is defined by state variables, which correspond to "genes" in a chromosome (the genotype). The atomic coordinates are the "phenotype". The "fitness" is the total interaction energy of the ligand with the protein. New individuals (offspring) are generated through crossover (inheriting genes from parents) and mutation (random changes to genes). Selection is based on fitness, allowing better-suited solutions to reproduce. Real-valued genes are used for ligand translation (three Cartesian coordinates), orientation (four variables defining a quaternion), and each torsion angle.

Quaternions are used for orientation to avoid "gimbal lock". New random number generators are introduced to ensure results can be reproduced on any hardware platform given the same seed values.

- (c) Lamarckian Genetic Algorithm (LGA) This is a hybrid search method combining the GA with a local search (LS) method. In LGA, environmental adaptations (local optimization by LS) of an individual's phenotype are "reverse transcribed" into its genotype and become heritable traits. This means the results of the local search update the individual's genotype, influencing future generations. The LS method is based on Solis and Wets, which has the advantage of not requiring gradient information about the local energy landscape, thus facilitating torsional space search. It is also adaptive, adjusting its step size based on success or failure in finding lower energies. The step size can be different for each type of gene (e.g., translational vs. rotational). LGA has shown enhanced performance relative to SA and GA alone. Across various test systems, LGA was found to be the most efficient (finding lowest energy for a given number of evaluations), most reliable (finding reproducible conformations), and most successful (reproducing crystal structures most often) among the three search methods. For instance, LGA successfully docked systems with up to 10 rotatable bonds where SA failed.

2. Exploration of Degrees of Freedom AutoDock's search algorithms explore a ligand's conformational space by manipulating its degrees of freedom:

- (a) Rigid Body Motion: The ligand can translate (move in X, Y, Z space) and rotate (change its orientation) as a rigid body. AutoDock uses a quaternion formalism for rigid body rotations to avoid "gimbal lock".
- (b) Torsional Flexibility: A key strength of AutoDock is its ability to handle flexible ligands with multiple torsional degrees of freedom. Each rotatable bond in the ligand is treated as a degree of freedom, allowing for changes in the ligand's conformation. The number of torsional degrees of freedom affects the total conformational entropy lost upon binding, which is accounted for in the force field.
- (c) Reflective Walls: The Cartesian search space is bounded by "reflective walls". If a ligand attempts to move outside the defined

grid volume, it is "bounced" back into the relevant simulation space by immediately giving it twice the opposite translational perturbation. This prevents the ligand from wasting simulation time exploring artificial surfaces at the grid edges.

- (d) Flexible Rings (Advanced): AutoDock can indirectly manage flexible rings by converting them to an acyclic (open) form and using special "G" atom types along with a 12-2 pseudo-Lennard-Jones potential to restore the cyclic structure during the calculation while exploring ring conformations. This potential guides the "edge atoms" next to each other, effecting ring closure. Multiple flexible rings can be docked simultaneously using different carbon atom types (G, J, Q, GA).
- (e) Protein Flexibility (Limited): While the protein is typically treated as a static target to enable rapid grid-based energy evaluation, AutoDock4 incorporates explicit conformational modeling of specified sidechains in the receptor. Selected sidechains can rotate around torsional degrees of freedom, similar to how the ligand's flexibility is treated. This feature is crucial for systems where receptor motion upon binding is significant.

3. User Input Minimization and Tools The researcher's input is minimized; they generally only need to specify the region of the protein containing the binding site and place the ligand in an arbitrary conformation and orientation nearby. The AutoDockTools (ADT) graphical user interface assists in preparing coordinate files (creating PDBQT files with polar hydrogens, partial charges, atom types, and torsional information), designing experiments (identifying active sites, setting search parameters), and analyzing results (clustering, visualization). ADT helps define rotatable bonds in the ligand and can even select specific protein sidechains to be treated as flexible. In essence, AutoDock searches for ligand conformations by performing numerous iterations of generating trial ligand positions and conformations (using SA, GA, or LGA), rapidly evaluating their energy with pre-computed grids, and then accepting or rejecting these trials based on the energy landscape and search algorithm's rules, ultimately aiming to find the lowest energy binding modes.

4. enumerate

The detailed scoring function that Autodock uses to score and rank the docking poses. AutoDock is designed to predict the bound conformations of

small, flexible ligands to macromolecular targets of known structure. Its core operation balances computational feasibility with a robust and physically relevant procedure. This involves efficient conformation searching coupled with rapid energy evaluation. AutoDock scores ligands after conformational search primarily using a semi-empirical free energy force field. This force field is designed to estimate the free energy of binding (ΔG) of small molecules to macromolecular targets.

1. Grid-Based Energy Evaluation (AutoGrid): Before the actual docking simulation, a separate program called AutoGrid pre-calculates three-dimensional "affinity grids". The target protein is embedded in this grid. A probe atom is sequentially placed at each grid point, and its interaction energy with the protein is calculated and stored at that point. Separate grids are generated for each type of atom present in the ligand (e.g., carbon, oxygen, nitrogen, hydrogen). Additionally, separate grids for electrostatic potential and desolvation potential are calculated. During the docking simulation, these grids act as rapid look-up tables. For each ligand atom, its interaction energy with the protein is determined by trilinear interpolation of the affinity values from the eight surrounding grid points. This makes the energy calculation very fast, as it's proportional only to the number of atoms in the ligand, not the protein. The calculation of these grids can incorporate various levels of sophistication, from constant dielectrics to finite difference methods.
2. Force Field Components:
 - The force field used by AutoDock is adapted from AMBER parameters. It accounts for various types of interactions:
 - (a) Dispersion/Repulsion: Modeled with a Lennard-Jones 12-6 potential.
 - (b) Hydrogen Bonding: Treated with a directional 12-10 potential. The parameters are assigned to yield a maximum well depth of -5 kcal/mol at 1.9 Å for hydrogen bonds with oxygen and nitrogen, and -1 kcal/mol at 2.5 Å for those with sulfur. A function $E(t)$ provides directionality based on the angle (t) from ideal hydrogen-bonding geometry.
 - (c) Electrostatics: Evaluated using a screened Coulomb potential. Earlier versions also used a constant dielectric of 40.
 - (d) Desolvation: Based on the volume of atoms that surround a given atom and shield it from solvent, weighted by a solvation parameter and an exponential term with a distance-weighting factor.

The desolvation model in AutoDock 4.2 is parameterized for all supported atom types, not just carbon.

3. Conformational Entropy: The force field also includes a term to estimate the unfavorable entropy lost upon ligand binding due to the restriction of its conformational degrees of freedom. This term is proportional to the number of rotatable bonds in the ligand (Ntor), excluding bonds that only move hydrogens.
4. Intramolecular Energies: Intramolecular interaction energies within the ligand (dispersion/repulsion, hydrogen bonding, electrostatic) are calculated at each time step. AutoDock 4.2’s default model for the unbound state assumes the ligand’s conformation is the same as its bound state, making the internal energy contribution zero. Earlier versions (like AutoDock 4.0) used an extended conformation for the unbound state, but this was changed due to problems with sterically-crowded molecules. Users can also specify a custom unbound energy.

The Scoring Function Equation AutoDock 4.2 uses a force field that evaluates binding in two steps: first, estimating the intramolecular energetics for the transition from the unbound states of the ligand and protein to their bound conformations, and second, evaluating the intermolecular energetics of combining them in their bound conformations. The general equation for the binding free energy (ΔG) is given as:

$$\begin{aligned} \Delta G = & \\ & (V_{\text{bound}}^{\text{L-L}} - V_{\text{unbound}}^{\text{L-L}}) + \\ & (V_{\text{bound}}^{\text{P-P}} - V_{\text{unbound}}^{\text{P-P}}) + \\ & (V_{\text{bound}}^{\text{P-L}} - V_{\text{unbound}}^{\text{P-L}}) + \\ & \Delta S_{\text{conf}} \end{aligned} \tag{2}$$

Where:

- L refers to the ligand.
- P refers to the protein.
- V^{L-L} represents the intramolecular interaction energy within the ligand.
- V^{P-P} represents the intramolecular interaction energy within the protein.

- V^{P-L} represents the intermolecular interaction energy between the protein and the ligand.
- ΔS_{conf} is the conformational entropy lost upon binding.

Each of the pairwise energetic terms (V) in the equation above includes evaluations for dispersion/repulsion, hydrogen bonding, electrostatics, and desolvation. These are combined with weighting constants (W) that have been optimized empirically based on a set of experimentally determined binding constants. The specific formulation of these pairwise terms is given as:

$$\begin{aligned}
 V = & \\
 & W_{vdw} \sum_{i,j} \left(\frac{A_{ij}}{r_{ij}^{12}} - \frac{B_{ij}}{r_{ij}^6} \right) + \\
 & W_{hbond} \sum_{i,j} E(t) \left(\frac{C_{ij}}{r_{ij}^{12}} - \frac{D_{ij}}{r_{ij}^{10}} \right) + \\
 & W_{elec} \sum_{i,j} \frac{q_i q_j}{\epsilon(r_{ij}) r_{ij}} + \\
 & W_{sol} \sum_{i,j} (S_i V_j + S_j V_i) e^{(-r_{ij}^2/\sigma^2)}
 \end{aligned} \tag{3}$$

Where:

1. W_{vdw} , W_{hbond} , W_{elec} , W_{sol} are the weighting constants determined through linear regression analysis. For example, in one model, dispersion/repulsion energies were weighted by 0.1485, electrostatics by 0.1146, and the desolvation term by 0.1711.
2. The first term is a Lennard-Jones 12-6 potential for dispersion/repulsion, where A_{ij} and B_{ij} are parameters, and r_{ij} is the distance between atoms i and j .
3. The second term is a directional 12-10 hydrogen bond potential, where C_{ij} and D_{ij} are parameters, and $E(t)$ provides directionality based on the angle t from ideal H-bonding geometry.
4. The third term is a screened Coulomb potential for electrostatics, where q_i and q_j are partial charges, and $\epsilon(r_{ij})$ is a distance-dependent dielectric function.
5. The final term is a desolvation potential based on the volume of atoms (V) that surround a given atom, weighted by a solvation parameter

(S) and an exponential term with a distance-weighting factor $\sigma = 3.5$ Å. The summation for this term in certain models, like Model C, is performed only over carbon atoms in the ligand.

Additionally, the torsional free energy (ΔG_{tor}) is included as a separate term, typically proportional to the number of rotatable bonds (N_{tor}) in the ligand, with a coefficient determined during calibration (e.g., 0.3113 kcal/mol per torsion). The full set of empirically determined coefficients for these terms was derived using linear regression against known binding constants from a set of protein-ligand complexes. After the conformational search, AutoDock performs a cluster analysis on the lowest energy docked structures found from multiple simulation runs. These clusters are then ranked by increasing energy. The software reports the fitness (docked energy), state variables, and coordinates of the docked conformation, along with the estimated free energy of binding (ΔG_{pred})

2.4 AUTODOCK-VINA

AutoDock Vina (Vina) employs a sophisticated method to sample ligand conformations, primarily through an efficient optimization algorithm that explores the ligand’s positional, orientational, and torsional flexibility. Here’s a breakdown of how AutoDock Vina samples ligand conformations:

1. Optimization Algorithm: Iterated Local Search (ILS)
 - (a) Vina utilizes an Iterated Local Search global optimizer, which is similar to an Iterated Monte-Carlo (MC) search algorithm. This was chosen after exploring various stochastic global optimization approaches like genetic algorithms, particle swarm optimization, and simulated annealing.
 - (b) The algorithm performs a succession of steps, each consisting of a mutation and a local optimization, with each step accepted according to the Metropolis criterion. This iterative process allows it to escape local minima and explore the conformational space more effectively.
 - (c) Multiple runs are performed, starting from random conformations. These runs can be executed concurrently using multithreading on multicore CPUs, leveraging shared-memory hardware parallelism to speed up the process. The algorithm maintains a set of diverse significant minima found during these separate runs, which

are then combined during the structure refinement and clustering stage. The number of steps in a run is determined adaptively based on the apparent complexity of the problem.

2. Local Optimization: BFGS Method with Gradients

- (a) For the local optimization steps within the Iterated Local Search, Vina uses the Broyden-Fletcher-Goldfarb-Shanno (BFGS) method. This is an efficient quasi-Newton method.
- (b) Crucially, BFGS uses not only the value of the scoring function but also its gradient. The gradient provides the derivatives of the scoring function for its arguments.
- (c) These arguments include the position and orientation of the ligand, as well as the values of the torsions for the active rotatable bonds in the ligand (and flexible residues, if any).
- (d) While evaluating the gradient in addition to the scoring function's value might take longer, using the gradient can significantly speed up the optimization process by giving the algorithm a "sense of direction".

- 3. Ligand Flexibility Docking generally assumes that the receptor is rigid, while a chosen set of covalent bonds in the ligand are considered freely rotatable (referred to as active rotatable bonds). Vina can handle ligands with a varying number of active rotatable bonds, from 0 to 32, as demonstrated in tests. The number of heavy atoms and active rotatable bonds can influence the time taken for a Vina run.
- 4. Scoring Function and Conformation Ranking The optimization algorithm aims to find the global minimum of the scoring function's conformation-dependent part and other low-scoring conformations, which it then ranks. The scoring function itself, while not directly sampling conformations, guides the optimization process to identify favorable binding poses.

5. Handling Search Space and Initial Conformations

- (a) Vina requires a user-defined "search space" in the receptor's coordinate system. For testing purposes, this space is typically expanded around the experimental ligand structure.
- (b) To prevent artificial help from a biased program, the initial ligand conformation (position, orientation, and torsions) is randomized

to be unrelated to the experimental structure, while also avoiding self-clashes.

6. New Features in Vina 1.2.0 Enhancing Sampling

- (a) **Macrocycle Conformational Sampling:** Vina 1.2.0 introduced a specialized protocol for docking macrocycles, which are challenging due to correlated torsional changes. This method involves breaking one bond in the ring structure, using dummy atoms, and applying a linear potential to restore the bond. This allows independent exploration of torsional degrees of freedom and sampling of macrocycle conformations while adapting to the binding pocket, although at the cost of increased search complexity.
- (b) **Control over Evaluations:** Users can now specify the number of energy evaluations to be performed using the " -- max_evals " option, providing more control over the search algorithm, particularly for complex searches (e.g., more flexible ligands or larger binding sites).

In summary, AutoDock Vina samples ligand conformations through an efficient Iterated Local Search, guided by a gradient-based BFGS local optimizer, which intelligently explores the ligand's translational, rotational, and torsional degrees of freedom. Recent updates further enhance its ability to handle complex flexibilities like those found in macrocycles.

Molecular docking, including the method used by AutoDock Vina, generally operates under the assumption that much or all of the receptor is rigid. This approach represents a trade-off where representational detail is reduced in favor of computational speed. Here's how AutoDock Vina treats receptor flexibility:

1. **Rigid Receptor Assumption:** The foundational principle for Vina, as with many docking programs, is that the receptor structure remains constant during the docking process. For example, in tests evaluating Vina's performance, the receptors were consistently treated as rigid, while ligands were allowed to be flexible.
2. **Gradient-Based Optimization:** While the primary focus of Vina's Broyden-Fletcher-Goldfarb-Shanno (BFGS) optimization method is on the ligand's position, orientation, and torsions, its arguments can also include the values of torsions for "flexible residues, if any". This indicates that Vina has a built-in capacity to account for some limited receptor flexibility (specifically, flexible residues) within its optimization algorithm,

although it’s not the default or primary mode of operation. The gradient calculation, which provides derivatives for these arguments, helps speed up the optimization process.

3. **Grid Maps and Post-processing:** Vina automatically calculates its own grid maps based on the target (receptor) structure. These maps are used to define intermolecular interactions. Furthermore, after docking, Vina uses the receptor structure to perform a post-processing minimization of the docked poses. This refinement step relies on direct pairwise interactions with the receptor. The consistency of these operations implies a fixed receptor structure being used for these calculations.
4. **User Control and Default Behavior:** Vina was designed with a limited amount of user input for performing docking, making it highly specialized and optimized. This design generally steers towards a rigid receptor assumption for simplicity and speed, as modifying receptor flexibility broadly would typically require significant changes in the source code.
5. **Comparison to AutoDock4:** AutoDock4 (AD4) allowed users to modify a larger number of docking parameters, providing more direct access to engine internals, which made it suitable for developing new docking methods. However, even with AD4, the concept of full receptor flexibility was complex. AutoDock Vina 1.2.0, while maintaining Vina’s core efficiency, introduced functionalities from the AutoDock Suite, some of which might implicitly or explicitly involve subtle receptor considerations (e.g., zinc coordination, which affects how the receptor interacts with specific ligand atoms), but the core assumption for bulk receptor movement generally remains rigid.

The scoring function used in AutoDock Vina (referred to as Vina) is designed to approximate the standard chemical potentials of the system, aiming to reproduce chemical potentials rather than directly dealing with energies. This approach is more akin to "machine learning" than purely physics-based, being justified by its performance on test problems. The general functional form of the conformation-dependent part of the scoring function in AutoDock Vina is given by:

$$c = \sum_{i < j} f_{t_i t_j}(r_{ij}) \quad (4)$$

- This summation is performed over all pairs of atoms that can move relative to each other, typically excluding 1–4 interactions (atoms separated by three consecutive covalent bonds).

- Each atom i is assigned a type t_i .
- $f_{t_i t_j}$ represents a symmetric set of interaction functions based on the interatomic distance r_{ij} . Hydrogen atoms are not explicitly considered in this sum, other than for atom typing. This value, c , can be further broken down into intermolecular and intramolecular contributions: $c = c_{\text{inter}} + c_{\text{intra}}$. The predicted free energy of binding (s_1) is calculated from the intermolecular part (c_{inter1}) of the lowest-scoring conformation:

$$\begin{aligned} s_1 &= g(c_1 - c_{\text{intra1}}) \\ &= g(c_{\text{inter1}}) \end{aligned} \quad (5)$$

- Here, c_1 refers to the total score of the lowest-scoring conformation, and c_{intra1} is its intramolecular contribution.
- The function g is an arbitrary, strictly increasing, smooth, and potentially nonlinear function. For other low-scoring conformations, s_i values are formally given to preserve ranking, using the intramolecular contribution of the best binding mode (c_{intra1}): $s_i = g(c_i - c_{\text{intra1}})$

The specific conformation-independent function g chosen for implementation is:

$$g(c_{\text{inter}}) = \frac{c_{\text{inter}}}{1 + wN_{\text{rot}}} \quad (6)$$

- N_{rot} represents the number of active rotatable bonds between heavy atoms in the ligand.
- w is the associated weight for N_{rot} . As per Table 1, the weight for N_{rot} is 0.0585. Components of the Interaction Functions ($f_{t_i t_j}$): The implementation of Vina’s scoring function was largely inspired by X-score and tuned using the PDBbind dataset. The interaction functions $f_{t_i t_j}$ are defined relative to the surface distance $d_{ij} = r_{ij} - R_{t_i} - R_{t_j}$, where R_t is the van der Waals radius of atom type t . In Vina’s scoring function, $h_{t_i t_j}$ (which $f_{t_i t_j}$ is equivalent to) is a weighted sum of steric interactions, hydrophobic interactions, and hydrogen bonding. The specific terms and their weights used in Vina’s scoring function are provided in Table 1:
- Steric Terms: These are identical for all atom pairs.
 - gauss1: weighted by -0.0356. $\text{gauss1}(d) = e^{-(d/0.5 \text{ \AA})^2}$
 - gauss2: weighted by -0.00516. $\text{gauss2}(d) = e^{-((d-3 \text{ \AA})/2 \text{ \AA})^2}$

$$- \text{Repulsion: weighted by 0.840. } \text{repulsion}(d) = \begin{cases} d^2, & \text{if } d < 0 \\ 0, & \text{if } d \geq 0 \end{cases}$$

- Hydrophobic Term: weighted by -0.0351. It equals 1 when $d < 0.5\text{\AA}$, 0 when $d > 1.5\text{\AA}$, and is linearly interpolated between these values.
- Hydrogen Bonding Term: weighted by -0.587. It equals 1 when $d < -0.7\text{\AA}$, 0 when $d > 0$, and is similarly linearly interpolated in between. Metals are formally treated as hydrogen bond donors.
- All interaction functions $f_{t_i t_j}$ are cut off at $r_{ij} = 8\text{\AA}$

Comparison with AutoDock4 (AD4) Scoring Function The Vina scoring function is quite different from the AD4 scoring function.

1. AD4 scoring function uses a physics-based model that includes van der Waals, electrostatic, and directional hydrogen-bond potentials derived from an early AMBER force field, along with a pairwise-additive desolvation term based on partial charges, and a simple conformational entropy penalty.
2. In contrast, Vina’s scoring function notably lacks explicit electrostatics and solvation terms. It relies on a van der Waals-like potential (combining repulsion and two attractive Gaussians), a non-directional hydrogen-bond term, a hydrophobic term, and a conformational entropy penalty.
3. The computational cost for energy evaluations is higher with the AD4 scoring function in Vina 1.2.0, taking nearly three times longer than with the Vina scoring function, due to the need to interpolate additional electrostatic and desolvation maps for each movable atom. Vina ranks conformations based on its scoring function, which includes both intermolecular and intramolecular contributions, differing from X-score, which only considers intermolecular contributions.

2.5 FLEXX

FLEXX performs ligand conformational search for docking by modelling ligand flexibility using a discrete model. This approach focuses on sampling the conformational space of the ligand efficiently. Here’s a breakdown of how it handles ligand flexibility:

1. Fixed Geometries: Bond lengths and angles are kept invariant as given in the input structure, except within ring systems. Therefore, the ligand structures provided to FLEXX should ideally be reasonably minimised geometries.
2. Acyclic Single Bond Flexibility (Torsion Angles):
 - (a) For acyclic single bonds, a set of preferred torsion angles is assigned.
 - (b) These preferred angles are determined by a statistical evaluation of the Cambridge Structural Database (CSD).
 - (c) The method uses a database of approximately 900 molecular fragments with a central single bond. By matching these representative torsional fragments onto the ligand, a histogram of torsion angle occurrences from the CSD is used to select highly populated torsion angles for generating low-energy conformations.
 - (d) Up to 12 discrete torsion angle values can be assigned to each acyclic single bond.
 - (e) Bonds adjacent to a methyl group or a planar amino group are exceptions and remain unchanged from the initial conformation.
3. Ring Flexibility:
 - (a) Multiple conformations for ring systems are computed using external programs.
 - (b) Initially, the SCA program (developed by De Clercq) was used, which computes alternative conformations for polycyclic systems containing five- to seven-membered rings. This was considered sufficient for most practical purposes in drug design and typically took only a few seconds for a typical drug.
 - (c) Later versions of FLEXX (e.g., version 1.6.5 evaluated in Kramer et al., 1999) used the CORINA program for computing multiple ring conformations, with a limit of seven ring atoms. Larger rings were treated as rigid based on the input structure.
4. Integration with Docking Algorithm: This discrete conformational model is fundamental to FLEXX’s incremental construction algorithm. The ligand is fragmented, and then fragments are added step by step in all possible conformations, allowing the system to explore the ligand’s flexible states as it builds the complex within the active site.

5. Accuracy and Complexity: The described model of ligand flexibility is reported to cause RMS deviations only in terms of conformational differences of less than 1 Å (e.g., 0.4 Å for methotrexate). While this model generates a combinatorial number of conformations, internal clashes discard a portion of these, providing an upper bound for the number of possible low-energy conformations. The ability to handle ligands with up to 17 rotatable bonds is sufficient for most practical drug design purposes. However, for ligands with more than 20 fragments, other search strategies might be required due to the heuristic greedy strategy employed during complex construction.
6. Pre-processing: Before docking, ligand structures are energy-minimised and assigned correct atom types, bond types, and formal charges to ensure suitable bond distances and angles and to remove any "docking information" from the PDB structure.

This systematic, discrete approach to ligand flexibility allows FlexX to efficiently sample a large variety of ligand conformations, aiming to reproduce the experimentally observed binding mode.

2.5.1 Scoring function

FLEXX scores the docked ligands using an empirical scoring function similar to that developed by Böhm (1994), with some minor changes. This function is designed to estimate the free binding energy (DG) of the protein-ligand complex. The general equation for the scoring function is: The equation for the scoring function is:

$$\begin{aligned}
 \Delta G = & \Delta G_0 + \Delta G_{\text{rot}} \times N_{\text{rot}} + \\
 & \Delta G_{\text{hb}} \sum_{\text{neutral H-bonds}} f(D_R, D_\alpha) + \\
 & \Delta G_{\text{io}} \sum_{\text{ionic int.}} f(D_R, D_\alpha) + \\
 & \Delta G_{\text{aro}} \sum_{\text{aro int.}} f(D_R, D_\alpha) + \\
 & \Delta G_{\text{lipo}} \sum_{\text{lipo. cont.}} f^*(D_R)
 \end{aligned} \tag{7}$$

Here's a breakdown of each term:

- ΔG_0 : This is a fixed ground term, set at 5.4 kJ/mol.

- $\Delta G_{\text{rot}} \times N_{\text{rot}}$: N_{rot} represents the number of free rotatable bonds in the ligand that become immobilized (hindered) upon binding to the complex. ΔG_{rot} accounts for the loss of entropy during ligand binding due to this immobilisation and is valued at 1.4 kJ/mol.
- $\Delta G_{\text{hb}} \sum_{\text{neutral H-bonds}} f(D_R, D_\alpha)$: This term is a sum over all pairwise neutral hydrogen bonds. $f(D_R, D_\alpha)$ is a scaling function that penalizes deviations from ideal geometry for these interactions. The ideal interaction distance for hydrogen bonds is 1.9 Å.
- $\Delta G_{\text{io}} \sum_{\text{ionic int.}} f(D_R, D_\alpha)$: This sums over all pairwise ionic interactions. $f(D_R, D_\alpha)$ is the same scaling function for geometric deviations. The ideal interaction distance for metal interactions is 2.0 Å.
- $\Delta G_{\text{aro}} \sum_{\text{aro int.}} f(D_R, D_\alpha)$: This term sums over aromatic interactions, specifically involving interactions between aromatic-ring-atoms, methyl, or amide groups with an aromatic-ring-centre. These interactions are used to generate the preferred t-shaped arrangement of neighbouring aromatic ring systems. In some versions, an additional parameter ΔG_{aro} of -0.7 kJ/mol is included. The ideal interaction distance for aromatic interactions is 4.5 Å.
- $\Delta G_{\text{lipo}} \sum_{\text{lipo. cont.}} f^*(D_R)$: This is a modification of Böhm’s original lipophilic contact energy, which was estimated using a grid method. In FLEXX, it’s calculated as a sum over all pairwise atom-atom contacts. $f^*(D_R)$ is a function that accounts for contacts with ideal distances and penalises forbiddingly close contacts. $D_R = R - R_0$, where R is the actual distance between atom centers and R_0 is its ideal value (sum of van-der-Waals radii, each increased by 0.3 Å). The function $f^*(D_R)$ has specific ranges for its values.

The model of protein-ligand interactions and the scoring function in FLEXX are derived from the work of Böhm (1992a, 1994) and Klebe (1994). The scoring function is computationally fast and easy to compute, contributing to FLEXX’s short runtimes. Its primary purpose is for ranking the generated docking solutions. While useful for distinguishing between good and bad inhibitors, the absolute binding energy values often show large differences (about 20 kJ/mol) from experimental values. This indicates that some important energetic contributions may not be fully modeled, and improvements to the scoring function are an ongoing area of development. For instance, it may not penalize large hydrophobic groups exposed to the solvent, potentially leading to misranking. Filter functions, which discard placements

with undesirable properties like insufficient steric complementarity or exposed lipophilic regions, are being developed to address these issues.

Key Aspects of Scoring in FLEXX:

- **Ranking Solutions:** The scoring function is primarily used for ranking the generated docking solutions.
- **Origin of Chemical Model:** The underlying chemical model in FLEXX is based on the work of Böhm (1992a, 1994) and Klebe & Mietzner (1994).
- **Interaction Types:** Interactions are classified by their geometric constraints and are used for placing the ligand. These include hydrogen bonds and some hydrophobic interactions (e.g., phenyl rings and methyl groups). These interactions are arranged on three levels, from highly directional (Level 3, e.g., H-bonds, metal acceptors) down to directionally unspecific (Level 1, e.g., aliphatic and aromatic carbon atoms, sulfur).
- **Heuristic Nature:** The functions f and f^* are described as heuristic distance and angle-dependent penalties.
- **Scoring Speed:** The scoring function is chosen to be fast and easy to compute, contributing to FLEXX's extremely short runtimes. Docking a typical drug molecule, considering ligand flexibility, can be performed in two to three minutes on a standard workstation, making it approximately ten times faster than alternative approaches for flexible ligand docking at the time of its development.
- **Limitations and Improvements:**
 - While useful for roughly distinguishing good and bad inhibitors, the absolute binding energy values often show large differences (about 20 kJ/mol) from experimental values.
 - The scoring function may not penalise large hydrophobic groups reaching out into the solvent, potentially leading to misranking of solutions.
 - Filter functions that extract placements with "undesirable properties" (e.g., insufficient steric complementarity or large lipophilic regions exposed to solvent) have been developed to improve ranking and quality.

- There is ongoing work to improve the scoring function to better model important energetic contributions that are currently not accounted for. Using a simple function based on a single configuration of a receptor-ligand complex for scoring is acknowledged as a rough approximation of the free energy of binding.
- Including water molecules explicitly is crucial for reproducing correct binding modes in some cases, which influences the interaction pattern and thus the score.

2.6 SURFLEX

Surflex-Dock is a widely used small-molecule docking program that combines an empirically parameterized scoring function with a surface-based similarity search strategy to generate and rank protein–ligand binding modes.[3, 4] It was originally introduced as Surflex,[3] implemented in the SYBYL environment, and has since been extended to Surflex-Dock 2.1 and beyond.[4]

Two key design principles underlie Surflex-Dock:

- An empirical scoring function that outputs values in units of pK_d , trained on experimental protein–ligand complexes and designed to be smooth, continuous, and piecewise differentiable for efficient gradient-based optimization.[4, 7]
- A molecular similarity-based search centered on an idealized protomol representation of the binding site, which allows rapid generation of candidate poses by aligning ligand fragments to this protomol.[3]

This document gives a conceptual overview of Surflex-Dock, its typical workflow, the theoretical basis of its scoring and search strategies, and a detailed explanation of the scoring function with equations suitable for training beginning users.

2.7 Search and optimization

Surflex-Dock 2.1 introduced several important enhancements to the search protocol:[4]

- **Quasi-Newton optimization:** Local pose optimization uses the BFGS quasi-Newton method with analytical gradients of the scoring function, improving both speed and robustness over simple steepest-descent approaches.

- **Small-molecule force field:** An implementation of the DREIDING force field is used to model intramolecular ligand energetics (bond lengths, angles, torsions, van der Waals interactions). This allows optimization of Cartesian coordinates beyond pure pose-parameter (translation, rotation, torsion) space.
- **Ring flexibility:** General dynamic exploration of flexible ring conformations (five-, six-, and seven-membered rings) is supported by mapping prototype ring conformers (e.g., cyclohexane) onto ligand ring templates and minimizing with the force field.
- **Fragment-guided docking:** When a particular substructure is known to bind in a defined manner (e.g., hinge-binding motifs in kinases, chelating groups in metalloenzymes), this fragment can be pre-placed and used to constrain and accelerate pose generation for analogs.

These extensions improve sampling of relevant poses, particularly for ligands with significant ring flexibility or initial conformational strain, and enhance screening robustness across diverse ligand sets.[4]

2.8 Scoring Function

Surflex-Dock uses an empirical scoring function that is constructed as a sum of terms corresponding to physically motivated contributions to binding:

- hydrophobic (steric) complementarity,
- polar (hydrogen-bond and electrostatic) complementarity,
- entropic penalties for fixing ligand degrees of freedom,
- solvation-related corrections.

Instead of computing an absolute binding free energy from first principles, the function is parameterized against a training set of protein–ligand complexes of known affinity and structure. The output is directly in units of

$$pK_d = -\log_{10}(K_d),$$

so that a one-unit change in the Surflex score roughly corresponds to an order-of-magnitude change in dissociation constant within the regime of the training data.[4, 7]

A notable methodological feature is the use of a multiple-instance learning framework during parameter estimation.[4] Rather than assuming that the exact crystallographic pose defines the score, the method:

1. Identifies a local optimum of the scoring function in the vicinity of the crystallographic pose.
2. Associates the complex’s experimental affinity with this locally optimal pose.

This reflects the idea that relatively small perturbations of a crystallographic pose are often experimentally indistinguishable, but can have large nominal effects on a given scoring function.

Subsequent work introduced the use of synthetically generated negative training data (physically unreasonable, interpenetrating poses) so that parameters controlling clash penalties and other “bad” interactions are estimated rigorously rather than heuristically.[7]

2.9 Protomol and molecular similarity-based search

A central concept in Surflex-Dock is the protomol.[3]

- The protomol is an idealized ligand-like object that occupies the binding site.
- It is constructed from small fragments such as methane (CH_4), amide carbonyl (C=O), and amide N-H , tessellated over the binding pocket and optimized using the Surflex scoring function itself.
- The protomol thus encodes the locations of preferred hydrophobic contacts, hydrogen bond donors, and hydrogen bond acceptors, as well as steric boundaries.

Docking a new ligand proceeds via surface-based molecular similarity:

1. Fragments of the input ligand are generated (e.g., by breaking rotatable bonds).
2. These fragments are aligned to the protomol using a rapid morphological similarity measure that compares molecular surfaces and chemical features.
3. High-similarity fragment poses are retained and locally optimized using the Surflex scoring function.
4. Compatible, high-scoring fragments are merged to generate full-ligand poses, which are further refined by gradient-based optimization.

Compared to purely grid-based or exhaustive sampling approaches, this strategy greatly reduces the search space by focusing on poses that resemble an idealized ligand for the pocket.

2.10 Typical Surflex-Dock Workflow

1. Step 1: Receptor preparation

(a) Select and clean the protein structure

- Choose an appropriate PDB structure (good resolution, relevant conformation, appropriate cofactors and metal ions).
- Remove crystallographic artifacts (e.g., irrelevant buffer molecules) while retaining essential structural waters, ions, and cofactors as needed.

(b) Add hydrogens and assign protonation states

- Add all hydrogens, not just polar hydrogens.
- Assign protonation and tautomeric states for ionizable residues and active-site residues based on pH and the known mechanism.

(c) Define the binding site

- If a co-crystallized ligand is present, define the binding site as a region around that ligand (e.g., within a chosen radius).
- Alternatively, define the site with a binding pocket-finding algorithm or by specifying residues.

2. Step 2: Protomol generation

Surflex can generate the protomol in two main ways:[3]

- **Ligand-based protomol:** When a co-crystallized ligand is available, the protomol is grown to mimic its interactions, essentially capturing an idealized version of that ligand's binding mode.
- **Pocket-based protomol:** When no ligand is available, the protomol can be constructed directly from the pocket geometry using fragment tessellation and optimization.

For training, a ligand-based protomol is often recommended because it usually yields more intuitive results and is easier to visualize.

3. Step 3: Ligand preparation

- (a) Generate 2D or 3D structures of ligands (from SMILES, SD files, etc.).
- (b) Standardize valence, formal charges, and tautomers as appropriate.

- (c) Assign protonation states consistent with the docking pH.
- (d) Optionally, pre-minimize ligands using the built-in DREIDING implementation to reduce conformational strain before docking.

4. Step 4: Docking setup and execution

Key Surflex-Dock options for beginners typically include:

- **Sampling level** (e.g., standard vs. thorough sampling).
- **Ring flexibility** (enable the ring-search option for ligands with flexible ring systems).
- **Fragment-guided docking** when a known binding substructure is available (using the appropriate fragment-matching options).
- **Number of poses per ligand** to be retained (e.g., top 10–20 poses).

The docking engine:

- (a) Aligns ligand fragments to the protomol using surface similarity.
- (b) Applies thresholds to eliminate gross steric clashes and low-similarity poses.
- (c) Merges compatible high-scoring fragments into full ligand poses.
- (d) Optimizes each pose using BFGS with respect to the Surflex scoring function.
- (e) Returns a ranked list of poses with associated Surflex scores (in units of pK_d).

5. Step 5: Post-processing and analysis

After docking:

- Inspect top-ranked poses visually for key interactions (hydrogen bonds, hydrophobic contacts, metal coordination).
- Cluster poses by root-mean-square deviation (RMSD) to identify distinct binding modes.
- Consider rescoring with alternative scoring functions or consensus scoring when prioritizing compounds for experimental testing.

3 The Surflex-Dock Scoring Function

3.1 Atomic surface distance

The Surflex-Dock scoring function is formulated in terms of *atomic surface distances* between protein and ligand atoms.[4]

For a ligand atom i and a protein atom j , define:

$$r_{ij} = d_{ij} - (R_i + R_j),$$

where:

- d_{ij} is the center-to-center distance between atoms i and j ,
- R_i and R_j are their van der Waals radii.

Thus:

- $r_{ij} > 0$ corresponds to separated atoms (no overlap),
- $r_{ij} = 0$ corresponds to exact contact of van der Waals surfaces,
- $r_{ij} < 0$ corresponds to interpenetration (steric clash or close contact).

Atoms are typed as *nonpolar* (e.g., carbon and attached hydrogens) or *polar* (e.g., heteroatoms involved in hydrogen bonding). Polar atoms may also carry formal charges.

3.2 Hydrophobic (steric) complementarity term

For each relevant atom pair, Surflex defines a steric (hydrophobic) complementarity score as a function of the surface distance r (writing r instead of r_{ij} for brevity):[4]

$$\text{steric_score}(r) = \lambda_1 \exp\left(-\frac{(r + n_1)^2}{n_2}\right) + \frac{\lambda_2}{1 + \exp(n_3(r + n_4))} + \lambda_3 \max(0, (r + n_5)^2). \quad (8)$$

Here:

- $\lambda_1, \lambda_2, \lambda_3$ and $n_1, \dots, n_5$ are empirically fitted parameters.
- The first term is a Gaussian centered near the optimal contact distance between hydrophobic atoms.

- The second term is a sigmoidal function controlling the mid-range attractive behavior.
- The third term is a quadratic penalty that activates for sufficiently positive r (i.e., when atoms are too close or clashing, depending on the shift n_5).

Physically, for ideal hydrophobic contacts, $\text{steric_score}(r)$ contributes on the order of 0.1 units of pK_d per favorable nonpolar atom–atom contact, with rapidly decaying contributions as the atoms move apart or interpenetrate excessively.[4]

3.3 Polar complementarity term

The polar complementarity term has a similar functional form but is applied to polar atom pairs and includes additional scaling to account for charge and directionality:[4]

$$\text{polar_score}(r) = \lambda_4 \exp\left(-\frac{(r + n_6)^2}{n_7}\right) + \frac{\lambda_5}{1 + \exp(n_3(r + n_8))} + \lambda_3 \max(0, (r + n_9)^2). \quad (9)$$

Key aspects:

- λ_4, λ_5 and $n_6, \dots, n_9$ are also empirically fitted.
- The parameter λ_3 is shared as the coefficient for the quadratic clash term in both hydrophobic and polar interactions, reflecting a common penalty model for interpenetration.
- The polar term is further scaled by:
 - a directionality factor that depends on the relative geometry (e.g., donor–hydrogen–acceptor angle and orientation),
 - a factor that depends quadratically on the formal charges of the interacting atoms.

For an ideal hydrogen bond (e.g., between a donor proton and a carbonyl oxygen), this term can contribute on the order of 1.2 units of pK_d , with an optimal surface separation corresponding to a typical donor–acceptor distance near 1.97 Å.[4]

3.4 Total Surflex-Dock score

The overall Surflex-Dock score for a given protein–ligand pose is a sum over pairwise atomic terms plus global entropic and solvation contributions:[4, 7]

$$S_{\text{Surflex}} = \sum_{(i,j) \in \mathcal{N}_{\text{np}}} \text{steric_score}(r_{ij}) + \sum_{(i,j) \in \mathcal{N}_{\text{pol}}} w_{ij}^{\text{pol}} \text{polar_score}(r_{ij}) + S_{\text{ent}} + S_{\text{solv}}. \quad (10)$$

In this expression:

- $\mathcal{N}_{\text{np}}$ is the set of nonpolar protein–ligand atom pairs that are considered for steric (hydrophobic) scoring.
- $\mathcal{N}_{\text{pol}}$ is the set of polar protein–ligand atom pairs.
- w_{ij}^{pol} is a weighting factor that encodes:
 - directional dependence of the polar interaction (e.g., hydrogen-bond angle),
 - scaling with formal charges of atoms i and j .
- S_{ent} represents entropic penalties:

$$S_{\text{ent}} = c_{\text{rot}} N_{\text{rot}} + c_{\text{tor}} N_{\text{tor}} + c_{\text{tr}} \log(\text{MW}), \quad (11)$$

where:

- N_{rot} is the number of rotatable bonds in the ligand,
- N_{tor} may count additional torsional contributions,
- MW is the molecular weight of the ligand,
- $c_{\text{rot}}, c_{\text{tor}}, c_{\text{tr}}$ are empirically estimated coefficients.
- S_{solv} is a solvation-related correction term, which in the original Surflex-Dock parameterization has relatively small weight but can be refined or customized.[4]

All coefficients and parameters (λ_k , n_k , and c_k) are obtained by fitting to training sets of protein–ligand complexes with known affinities using the multiple-instance learning protocol. Later refinements incorporate negative training data (unphysical poses) to better constrain the parameters governing interpenetration and other unfavorable interactions.[7]

3.5 Interpretation of Surflex scores

Because S_{Surflex} is in units of pK_d ,

- A higher score indicates stronger predicted binding affinity.
- Differences in score can be roughly interpreted as differences in log affinity, e.g., a difference of approximately 1 unit in score corresponds to roughly a tenfold change in predicted K_d , within the limits of the training set and the empirical model.

For virtual screening applications, absolute values of S_{Surflex} are typically less important than *relative* scores across a congeneric series or a screening library.[4, 7]

3.6 GOLD

GOLD (Genetic Optimisation for Ligand Docking) is a widely used protein–ligand docking program that uses a genetic algorithm (GA) to explore ligand binding modes in a (partially) flexible protein binding site.[5] It is developed and distributed by the Cambridge Crystallographic Data Centre (CCDC).

GOLD is particularly known for:

- robust pose prediction with full ligand flexibility and optional side-chain and water flexibility;
- multiple scoring functions (GoldScore, ChemScore, ASP, ChemPLP) and consensus scoring;[8, 6]
- explicit handling of binding-site water molecules, including displacement and reorientation;
- highly configurable constraints (distance, H-bond, pharmacophore, scaffold, etc.) to encode prior knowledge.

GOLD’s design is typical of modern docking programs: a stochastic search procedure (here, a GA) generates candidate poses, and an empirical or knowledge-based scoring function evaluates their fitness.[5] The combination of a flexible GA and several complementary scoring functions makes GOLD a good teaching platform for beginners.

3.7 Fundamental working theory of GOLD

In structure-based docking, the goal is to predict the bound pose of a ligand in a protein binding site, and, ideally, to estimate its binding affinity. Conceptually, one must:

1. sample the relevant conformational and orientational space of the ligand (and sometimes the receptor);
2. evaluate the quality of each sampled pose with a scoring function that approximates the binding free energy.

The search space is high-dimensional: it includes ligand translations and rotations, rotatable bonds, ring conformations, and, in GOLD, possible flips of certain protein side chains and on/off states and orientations of some water molecules.[5]

3.7.1 Genetic algorithm in GOLD

GOLD uses a steady-state, operator-based genetic algorithm to perform the conformational search.[5] A genetic algorithm is an evolutionary optimisation method that maintains a population of candidate solutions (“chromosomes”) and iteratively improves them through selection, crossover, mutation, and (optionally) migration between subpopulations (islands).

Encoding (chromosomes). In GOLD, each chromosome encodes a complete ligand pose:[5]

- the three translational degrees of freedom of the ligand;
- three rotational degrees of freedom (e.g. Euler angles or quaternions);
- values for each rotatable bond in the ligand;
- optional discrete variables for ring conformations;
- optional side-chain torsion states for flexible protein residues near the binding site;
- optional on/off (bound/displaced) flags and orientations for selected water molecules.

Each chromosome thus represents a complete candidate docking solution.

Fitness evaluation. The fitness (objective) function is a docking score: GoldScore, ChemScore, ASP, or ChemPLP. Higher scores correspond to better predicted binding (GOLD reports a dimensionless fitness function rather than an energy in physical units).

GA cycle. The GA proceeds as follows:[5]

1. **Initialisation:** generate a random or biased initial population of chromosomes obeying simple steric constraints.
2. **Evaluation:** compute the fitness score for each chromosome.
3. **Selection:** select parent chromosomes with probability biased towards higher fitness (e.g. tournament or roulette-wheel selection).
4. **Variation:** apply genetic operators:
 - *crossover* — recombine two parents to produce offspring (e.g. swapping subsets of torsions or pose parameters);
 - *mutation* — randomly perturb a degree of freedom (e.g. rotate one torsion, translate or rotate the ligand slightly, flip a side chain).
5. **Replacement (steady state):** insert offspring into the population, replacing some of the less fit chromosomes, often with duplicate suppression so that identical solutions are not preserved.[5]
6. **Termination:** after a fixed number of genetic operations or when convergence criteria (e.g. several best poses within a small RMSD) are met, the GA stops and returns the highest scoring poses.

GOLD uses an island model, where the population is partitioned into islands that evolve semi-independently with occasional migration. This encourages diversity and helps avoid premature convergence.[5]

3.8 Protein and water flexibility

While the receptor is usually treated as rigid, GOLD supports limited protein and water flexibility:

- **Protein side chains:** selected residues in the binding site can be treated as flexible, typically restricted to a library of side-chain rotamers or sampled torsions.
- **Water molecules:** explicitly modelled waters can be allowed to:

- toggle between bound (in the site) and displaced (returned to bulk);
- rotate to optimise hydrogen bonding;
- (optionally) translate within a small radius.

GOLD estimates the free-energy change associated with bringing a water molecule from bulk into a particular site, and this contributes to the fitness of poses where that water is retained versus displaced.

4 Typical GOLD workflow for beginners

A standard GOLD workflow can be broken into the following stages.

4.1 Protein (receptor) preparation

1. **Start from a high-quality structure.** Use a crystal structure or cryo-EM structure with a bound ligand when possible.
2. **Clean the structure:** remove alternate conformations, resolve missing atoms if possible, and inspect for crystallographic artefacts.
3. **Add hydrogen atoms:** add all hydrogens and choose appropriate protonation/tautomeric states for titratable residues given the target pH.
4. **Handle cofactors and metals:** retain essential metal ions and prosthetic groups; ensure their coordination geometry is reasonable.
5. **Water molecules:**
 - keep structural waters that are known or suspected to be functionally important;
 - remove obvious bulk waters;
 - flag selected waters as potentially “toggleable” in GOLD for binding/displacement analysis.

4.2 Binding-site definition

The binding site can be defined in several ways:

- **From a reference ligand:** specify a co-crystallised ligand; GOLD then defines a spherical binding site of user-defined radius (e.g. 8–12 Å) around the ligand.
- **By residues:** specify a set of binding-site residues; GOLD uses their coordinates to define the pocket region.
- **By coordinates:** specify a point and radius (useful when docking into apo structures or homology models).

For training purposes, defining the site around a known ligand is recommended, as it provides a clear structural context and an RMSD reference for pose validation.

4.3 Ligand preparation

1. **2D to 3D:** build ligands or import from databases and generate 3D structures.
2. **Protonation and tautomers:** enumerate plausible protomers and tautomers at the working pH; many failures in docking arise from incorrect protonation states.
3. **Conformation:** optionally pre-minimise ligands with a small-molecule force field; GOLD will sample conformations during docking but a reasonable starting structure can help.
4. **Define rotatable bonds:** GOLD automatically detects rotatable bonds, but manual curation (e.g. ring amides, conjugated bonds) is sometimes necessary.

4.4 Docking setup in GOLD / Hermes

- **Choose the scoring function:** ChemPLP is the default and generally recommended for pose prediction and virtual screening.
- **GA parameters:** select:
 - population size,
 - number of genetic operations,
 - selection pressure,
 - operator weights (crossover, mutation, migration).

The search efficiency parameter in recent versions provides a convenient, ligand-size-aware way to scale the number of GA operations.

- **Number of GA runs per ligand:** for pose prediction, 10–30 independent runs per ligand are typical; for virtual screening, fewer runs may be used for speed.
- **Constraints (optional but powerful):**
 - enforce key hydrogen bonds or metal interactions;
 - require a pharmacophore pattern or a scaffold match;
 - apply distance or shape similarity constraints relative to a reference ligand.

4.5 Running docking and analysing results

After docking:

1. **Rank poses** by the chosen fitness function.
2. **Inspect top poses** visually for:
 - agreement with known SAR and chemistry;
 - reasonable geometry (no severe clashes, realistic torsions);
 - preservation or rational modification of key interactions.
3. **RMSD analysis:** for systems with known co-crystal structures, compute RMSD between predicted and experimental poses ($\text{RMSD} \lesssim 2.0 \text{ \AA}$ is commonly considered a successful pose).[5]
4. **Rescoring / consensus scoring:** re-score docked poses with alternative scoring functions (e.g. dock with ChemPLP, rescore with GoldScore and ChemScore) to reduce scoring-function-specific bias.[8]

5 Scoring functions in GOLD

GOLD provides four main scoring functions:

- GoldScore (force-field-like, early default);
- ChemScore (empirical, trained against binding affinities);
- ASP (Astex Statistical Potential, knowledge-based);[6]

- ChemPLP (empirical, piecewise linear potential, current default).

All are dimensionless “fitness” scores; higher values indicate better predicted binding.

5.1 GoldScore

GoldScore is the original GOLD scoring function and is designed primarily for pose prediction.[5] Conceptually, it is a force-field-inspired function consisting of:

- an external van der Waals term between protein and ligand;
- an external hydrogen-bonding term between protein and ligand;
- an internal ligand strain term (van der Waals and torsional components).

In the implementation used in docking benchmarks, GoldScore is typically written as:[9]

$$S_{\text{GoldScore}} = 1.375 S_{\text{vdw,ext}} + S_{\text{hb,ext}} + S_{\text{int}}, \quad (12)$$

where:

- $S_{\text{vdw,ext}}$ is the van der Waals steric interaction between protein and ligand (4–8 potential);[9]
- $S_{\text{hb,ext}}$ is the hydrogen-bonding interaction between protein and ligand;
- S_{int} is an internal ligand strain term based on van der Waals and torsional contributions.

The empirical factor 1.375 partially compensates for the absence of an explicit hydrophobic term.[9]

The external van der Waals term is computed as a sum over all protein–ligand atom pairs (i, j) :

$$S_{\text{vdw,ext}} = \sum_{i \in L} \sum_{j \in P} E_{ij}^{\text{vdw}}(r_{ij}), \quad (13)$$

where r_{ij} is the interatomic distance. A 4–8 Lennard–Jones-like potential is used for the external van der Waals interactions:

$$E_{ij}^{\text{vdw}}(r_{ij}) = \epsilon_{ij} \left[\left(\frac{r_{ij,0}}{r_{ij}} \right)^8 - 2 \left(\frac{r_{ij,0}}{r_{ij}} \right)^4 \right], \quad (14)$$

where:

- $r_{ij,0}$ is a distance parameter related to the sum of van der Waals radii of atoms i and j ;
- ϵ_{ij} is an interaction well depth parameter depending on the atom types and interaction class (e.g. hydrogen-bonding vs. non-polar contact).

The hydrogen-bond term is a distance- and angle-dependent potential:[5]

$$S_{\text{hb,ext}} = \sum_{\text{HB pairs}} E_{ij}^{\text{hb}}(r_{ij}, \theta_{ij}), \quad (15)$$

where E_{ij}^{hb} is maximally favourable near an ideal donor–acceptor distance and angle, and decays as the geometry deviates from ideal hydrogen bonding.

Finally, the internal term S_{int} penalises high-energy ligand conformations (steric clashes, unfavourable torsions), discouraging poses that require unrealistic ligand strain.

5.2 ChemScore

ChemScore is an empirical scoring function originally developed by Eldridge *et al.* for estimating protein–ligand binding free energies from structures.[1] It was later implemented and adapted in GOLD.[8] In its original form, ChemScore estimates the binding free energy as:

$$\Delta G_{\text{bind}} = \Delta G_0 + \Delta G_{\text{hbond}} \sum_i H_i + \Delta G_{\text{metal}} \sum_j M_j + \Delta G_{\text{lipo}} \sum_k L_k + \Delta G_{\text{rot}} N_{\text{rot}}, \quad (16)$$

where:

- ΔG_0 is a constant offset (overall regression intercept);
- $\sum_i H_i$ is a sum over hydrogen-bonding interactions, each weighted by a geometry-dependent factor and the strength of the donor/acceptor;
- $\sum_j M_j$ is a sum over metal–ligand interactions (for ligands coordinating metal ions);
- $\sum_k L_k$ is a measure of lipophilic contact area (buried hydrophobic surface or hydrophobic contact count);
- N_{rot} is the number of rotatable bonds in the ligand in the unbound state;

- ΔG_{hbond} , ΔG_{metal} , ΔG_{lipo} , and ΔG_{rot} are empirical coefficients obtained by regression against a training set of 82 complexes with known binding affinities.[1]

In GOLD, the ChemScore-based fitness[9] incorporates additional penalties:

$$S_{\text{ChemScore}}^{\text{GOLD}} = -\Delta G_{\text{bind}} + P_{\text{clash}} + P_{\text{torsion,int}}, \quad (17)$$

where:

- P_{clash} penalises protein–ligand atom clashes;
- $P_{\text{torsion,int}}$ is an internal torsion penalty discouraging energetically unfavourable ligand conformations (complementary to N_{rot}).

A higher $S_{\text{ChemScore}}^{\text{GOLD}}$ corresponds to a more favourable (more negative) ΔG_{bind} and fewer penalties.

5.3 ASP: Astex Statistical Potential

ASP is a knowledge-based, atom–atom distance potential derived from a database of protein–ligand complexes.[6] It is conceptually related to PMF and DrugScore potentials but uses an improved reference state that accounts for differences in exposure of protein atom types towards ligand binding sites.[6]

In general, a statistical potential for atom types i and j at separation r can be written as:

$$w_{ij}(r) = -k_{\text{B}}T \ln \left(\frac{\rho_{ij}^{\text{obs}}(r)}{\rho_{ij}^{\text{ref}}(r)} \right), \quad (18)$$

where:

- k_{B} is Boltzmann’s constant;
- T is an effective temperature (often absorbed into the scale of the potential);
- $\rho_{ij}^{\text{obs}}(r)$ is the observed distribution of i – j distances in a structural database of complexes;
- $\rho_{ij}^{\text{ref}}(r)$ is the reference (non-interacting) distribution that corrects for background atom exposure and other biases.

The ASP score for a pose is then:

$$S_{\text{ASP}} = \sum_{i \in P} \sum_{j \in L} w_{ij}(r_{ij}) + S_{\text{ChemScore,aux}}, \quad (19)$$

where $S_{\text{ChemScore,aux}}$ denotes additional ChemScore-like terms (e.g. for hydrogen bonding and rotational entropy) that are incorporated into the ASP fitness in GOLD.

Because ASP is knowledge-based, it captures typical contact preferences (e.g. aromatic stacking, specific polar contacts) encoded in the database. It often performs comparably to GoldScore and ChemScore for pose prediction and binding-affinity ranking.[6]

5.4 ChemPLP: piecewise linear potential

ChemPLP is the current default scoring function in GOLD and is optimised for pose prediction and virtual screening. It combines:

- a piecewise linear potential (PLP) for steric (van der Waals-like) protein–ligand interactions;
- ChemScore-like hydrogen-bond and metal-binding terms;
- internal ligand clash and torsion terms.

In schematic form, the ChemPLP score can be written as:

$$S_{\text{ChemPLP}} = \sum_{i \in L} \sum_{j \in P} f_{\text{PLP}}(r_{ij}) + \sum_{\text{HB}} f_{\text{HB}}(r, \theta) + S_{\text{lig,clash}} + S_{\text{torsion}}, \quad (20)$$

where:

- $f_{\text{PLP}}(r_{ij})$ is a piecewise linear approximation to a steric potential, with different linear segments modelling strong repulsion at short range, optimal contact, and decay at longer range;
- $f_{\text{HB}}(r, \theta)$ is a distance- and angle-dependent hydrogen-bond term derived from ChemScore;
- $S_{\text{lig,clash}}$ penalises internal heavy-atom clashes within the ligand;
- S_{torsion} is a torsional strain term similar to that used in ChemScore.

The details of the breakpoints and slopes in $f_{\text{PLP}}(r)$ are proprietary, but conceptually the PLP term rewards close, well-packed contacts while sharply penalising severe overlaps. Validation studies by CCDC and others have shown ChemPLP to provide the highest success rates for pose prediction among the GOLD scoring functions, which is why it is the default choice in GOLD 5.1 and later.

6 Choosing and using scoring functions in training

For teaching beginners, the following practical guidelines are useful:

- Use **ChemPLP** as the default scoring function for pose prediction and small to medium virtual screening tasks.
- Demonstrate **GoldScore** to illustrate a more force-field-like potential and discuss its energy components explicitly.
- Use **ChemScore** when emphasising empirical free-energy decomposition into hydrogen bonding, lipophilic, metal, and rotational terms, and stress that absolute values are not true physical ΔG .
- Present **ASP** to introduce knowledge-based potentials and how they are derived from structural statistics.
- Encourage **consensus scoring** (e.g. dock with ChemPLP, rescore top poses with GoldScore and ChemScore) to highlight the strengths and weaknesses of different scoring functions and to reduce dependence on any single model.[8]

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